

一. 是非题

是 是 非 非 是 是 是 .

二. 选择题

(b) (a) (b) (d) (c) .

三. 填空题 (18 分, 每题 3 分) [方括弧内为 B 卷答案]

1. 4/7 . 2. $f_Y(y) = \begin{cases} 1/(4\sqrt{y}) & 0 < y < 4 \\ 0 & \text{其他} \end{cases}$
 3. 0.8446 . 4. 0.1 . 5. 1/3; 2 . 6. 上限为 6.356 .

四. 计算与应用题

1. A —— 任取 2 箱都是民用口罩,

B_k —— 丢失的一箱为 k $k = 1, 2, 3$ 分别表示民用口罩, 医用口罩, 消毒棉花.

$$P(A) = \sum_{k=1}^3 P(B_k)P(A|B_k) = \frac{1}{2} \cdot \frac{C_4^2}{C_9^2} + \frac{3}{10} \cdot \frac{C_5^2}{C_9^2} + \frac{1}{5} \cdot \frac{C_5^2}{C_9^2} = \frac{8}{36}$$

$$P(B_1|A) = P(B_1)P(A|B_1)/P(A) = \frac{1}{2} \cdot \frac{C_4^2}{C_9^2} / P(A) = \frac{3}{36} \div \frac{8}{36} = \frac{3}{8}.$$

2. (1) $A = 1/4$.

$$(2) f_X(x) = \int_{-\infty}^{\infty} f(x, y)dy = \begin{cases} \int_{-x}^x (1/4)dy = x/2 & 0 < x < 2 \\ 0 & \text{其他} \end{cases}$$

$$\text{当 } 0 < x < 2 \text{ 时, } f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} 1/(2x) & -x < y < x \\ 0 & \text{其他} \end{cases}$$

$$(3) E(X) = \int_0^2 (x^2/2)dx = 4/3, \quad E(Y) = \int_0^2 dx \int_{-x}^x (y/4)dy = 0,$$

$$E(XY) = \int_0^2 xdx \int_{-x}^x (y/4)dy = 0, \quad \text{cov}(X, Y) = E(XY) - E(X)E(Y) = 0$$

所以 X 与 Y 不相关.

$$3. f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{其他} \end{cases} \quad f_Y(y) = \begin{cases} e^{-y} & y \geq 0 \\ 0 & y < 0 \end{cases}$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(2x-z)dx$$

$$\begin{cases} 0 < x < 1 \\ 2x - z > 0 \end{cases} \Rightarrow \begin{cases} 0 < x < 1 \\ x > z/2 \end{cases} \Rightarrow \text{得 } z \text{ 轴上的分界点 } 0 \text{ 与 } 2$$

$$f_Z(z) = \begin{cases} \int_0^1 e^{-(z-2x)} dx = e^z(1-e^{-2})/2 & z \leq 0 \\ \int_{z/2}^1 e^{-(z-2x)} dx = (1-e^{z-2})/2 & 0 < z < 2 \\ 0 & z \geq 2 \end{cases}$$

4. 设 $X_i = \begin{cases} 1 & \text{第 } i \text{ 台彩电为次品且未被查出} \\ 0 & \text{其他} \end{cases} \quad i = 1 \sim 2 \times 10^5$

$$E(X_i) = 5 \times 10^{-6}, \quad D(X_i) = 5 \times 10^{-6}(1 - 5 \times 10^{-6})$$

$$\text{经检验后的次品数 } Y = \sum_{i=1}^{2 \times 10^5} X_i, \quad E(Y) = 1, \quad D(Y) = 1 - 5 \times 10^{-6},$$

$$\text{由中心极限定理, 近似地有 } Y \sim N(1, 1 - 5 \times 10^{-6})$$

$$P(Y > 3) = 1 - P(Y \leq 3) \approx 1 - \Phi\left(\frac{3-1}{\sqrt{1-5 \times 10^{-6}}}\right) \approx 1 - \Phi(2) = 0.0228.$$

5. (1) $\bar{X} = \sum_{i=1}^8 X_i = 16/8 = 2$, 令 $E(X) = 3 - 4p = \bar{X}$,

$$\text{得 } p \text{ 的矩估计为 } \hat{p} = (3 - \bar{X})/4 = 1/4.$$

(2) 似然函数为

$$\begin{aligned} L(p) &= \prod_{i=1}^8 P(X = x_i) = P(X=0)[P(X=1)]^2 P(X=2)[P(X=3)]^4 \\ &= 4p(1-p)^2(1-2p)^4 \end{aligned}$$

$$\ln L(p) = \ln 4 + 6 \ln p + 2 \ln(1-p) + 4 \ln(1-2p)$$

$$\text{令 } [\ln L(p)]' = \frac{6}{p} - \frac{2}{1-p} - \frac{8}{1-2p} = 0, \quad \Rightarrow 12p^2 - 14p + 3 = 0$$

$$\Rightarrow p = (7 \pm \sqrt{13})/12. \text{ 由 } 0 < p < 1/2, \text{ 故 } p = (7 + \sqrt{13})/12 \text{ 舍去}$$

$$\text{所以 } p \text{ 的极大似然估计值为 } \hat{p} = (7 - \sqrt{13})/12 = 0.2828.$$

6. 由样本得 $\bar{X} = 1267$, $S = \sqrt{\frac{1}{3} \sum_{i=1}^4 (X_i - \bar{X})^2} = \sqrt{40/3} = 3.65.$

(1) 要检验的假设为 $H_0: \mu = 1260, H_1: \mu \neq 1260$)

检验用的统计量 $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$,

拒绝域为 $|T| \geq t_{\frac{\alpha}{2}}(n-1) = t_{0.025}(3) = 3.1824$.

$|T_0| = \frac{1267 - 1260}{3.65/\sqrt{4}} = 3.836 > 3.1824$, 落在拒绝域内,

故拒绝原假设 H_0 , 即不能认为结果符合公布的数字 1260°C .

(2) 要检验的假设为 $H_0: \sigma \leq 2, H_1: \sigma > 2$

检验用的统计量 $\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$,

拒绝域为 $\chi^2 > \chi_{\alpha}^2(n-1) = \chi_{0.05}^2(3) = 7.815$

$\chi_0^2 = 40/4 = 10 > 7.815$, 落在拒绝域内,

故拒绝原假设 H_0 , 即不能认为测定值的标准差不超过 2°C .

五、证明题 (6分)

由题设 $P(X=m) = \frac{3^m}{m!}e^{-3}$, $P(Y=n) = \frac{3^n}{n!}e^{-3}$, $n, m = 0, 1, 2, \dots$

$$\begin{aligned} P(X+Y=i) &= \sum_{k=0}^i P(X=k, Y=i-k) = \sum_{k=0}^i P(X=k)P(Y=i-k) \\ &= \sum_{k=0}^i \frac{3^k}{k!}e^{-3} \cdot \frac{3^{i-k}}{(i-k)!}e^{-3} = e^{-6} \frac{1}{i!} \sum_{k=0}^i \frac{i!}{k!(i-k)!} 3^k \cdot 3^{i-k} \\ &= e^{-6} \frac{1}{i!} (3+3)^i = \frac{6^i}{i!} e^{-6}, \quad i = 0, 1, 2, \dots \end{aligned}$$

所以 $X+Y$ 仍服从泊松分布, 参数为 6.